How to Sketch a Graph of the function $y = f(x)$:

1) Find the _____ (the set of all x-values for which $f(x)$ is defined).
   Determine whether the function is even, odd, or neither.
   If the function is even, it is symmetric about the _____.
   If the function is odd, it is symmetric about the _____.
   If the function is neither, _____.

2) Find the _____ and _____ derivatives.

3) Find the _____ _____ by setting _____ equal to zero and finding where _____ is undefined. Use the _____ or _____ Derivative tests to find local extrema.

4) Determine where the function is _____ and _____.
   If $f'(x) > 0$ on some interval (a,b) then $f(x)$ is _____.
   If $f'(x) < 0$ on some interval (a,b) then $f(x)$ is _____.

5) Find any _____ _____ by setting _____ equal to zero and finding where _____ is undefined. Be sure to check that $f''(x)$ _____ _____ !!!
   State the intervals where $f(x)$ is _____ up or _____ down.

6) Find any vertical, horizontal, or oblique _____.
   a) To find vertical _____ we take the _____ as x approaches a from the _____ and as x approaches a from the _____ for any values of a that make the original function _____.
   b) If the degrees of the numerator and denominator of $f(x)$ are the _____ OR if the degree of the denominator is strictly _____ than the degree of the numerator, there is a _____ _____.
   c) If the degree of the numerator is exactly _____ _____ than the degree of the denominator, there is an _____ _____.
   To find the asymptotes in parts (b) and (c), we take the _____ as x approaches _____.

7) Plot any key points such as the _____ and _____ intercepts and any points found in steps 1-5. Be sure to plug the x-coordinates back into the _____ function to find the y-coordinates. Plot any _____ found in Step 6. Combine the information gathered in steps 1-6 to sketch the curve.